

YIPING YIN

Market Microstructure | Asset Pricing & Forecasting | Multi-Agent Market Simulation

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EDUCATION

UNSW Sydney | Bachelor of Commerce (Finance Major) | Financial Engineering Plan Sep 2023 - Sep 2026
Optiver x UNSW | Data and Algorithms in Trading | **High Distinction**
UNSW International Student Award | **Receive 20% toward tuition fees** of the program

COMPETITIONS

Algothon | Susquehanna International Group (SIG) 2026
Forecasting Under a Limited Effective Sample. General: **1st/253**; Final Technical: **3rd**; **69-page report**.
BigAlpha | BigQuant x Ingram Micro 2026
Latent-Agent Market Simulation. Score: **87**; **Top 3** in AI Open Innovation.

RESEARCH EXPERIENCE

UNSW Sydney | Research Assistant | Supervisor: Dr Xianghua Wu Jun 2026 - Present
• Analyzing a large-scale agent-society dataset for AI-Agent Society & Strategic Simulation research, examining how shared-model bias, memory, information, incentives and network position shape collective outcomes.
• Drafted probe-agent, controlled-intervention and external-model counterfactual protocols; defined evaluation measures for diffusion speed, network concentration, knowledge correction and governance response.

QUANTITATIVE PROJECTS

Limit-Order-Book Analytics, ETF Pricing & Market-Making Research Feb 2025 - May 2025
• Reconciled **4,406 PXA trades to 24,303 eligible L1-L8 order-book states** with same-venue strict-prior matching; benchmarked four mid-price estimators and generated 34,044 rolling-100 diagnostics with zero look-ahead rows.
• Built a five-second NDQ fair-value/iNAV model from 4,319 synchronized ETF, NQ futures and AUD/USD observations; the beta-1 specification recorded **4.30 bps MAE and 0.999 correlation** on 4,259 post-calibration rows.
• Designed a 64-session chronological minute-bar-proxy market-making study; on the 12-session historical diagnostic partition, inventory/imbalance controls reduced fills from **1,863 to 1,014** and maximum inventory from **500 to 336** while diagnostic net P&L rose from 56.53 to 147.21 under identical proxy assumptions.

Market Microstructure & Cross-Asset Market Making | Optiver x UNSW May 2026 - Aug 2026
• Analyzed **1,003 timestamp-matched OB5X futures-ETF order-book observations** from a 20-minute Optibook session; found negligible lead-lag within a three-second window and return correlation rising from **0.032 at 1 second to 0.522 at 120 seconds**, consistent with an Epps-effect pattern.
• Built a cross-asset market-making system that normalized the OB5X five-stock basket, three futures and ETF into common index units, executed depth-matched ETF-futures IOC pairs, and quoted two Black-76 options with inventory-aware skew and threshold-based delta hedging.
• Completed a one-hour Optibook simulation with **+36.50 P&L change on 1,222 lots and 187 fills**; 39 arbitrage pairs matched 892 lots, all 6,972 quote cycles remained two-sided, maximum |delta| was 7.51, and cleanup ended flat with zero loop errors, reconnects or self-trades.

Signal Mosaic - Calendar-Aligned Multi-Asset & News Analytics Jul 2026 - Present
• Engineered a reproducible 2020-2023 foundation spanning **50 US equities, 10 cryptocurrencies and 149,683 raw headlines**; computed native-calendar returns and aligned news to the same or next equity trading day.
• **Removed 2,847 exact duplicates** and excluded 10 post-sample crypto rows; built breadth and regime diagnostics with point-in-time availability and lagged-sentiment controls.

Explaining Markets - Earnings-Reaction Forecasting | Chicago Booth X Optiver Aug 2026 - Present
• Built and froze a **50,000-feature TF-IDF/Ridge residual model** on **6,147 asset-event examples** (Q4 2025-Q2 2026), using cutoff-bounded official disclosures, quarter-wise earnings-surprise residualization and a fixed forward scheme with **2,000 source-event block bootstraps** under the pinned official delta R-squared scorer.
• Engineered a five-minute signed ACK-first submission path with atomic deduplication, post-ACK asynchronous inference and exact-neutral fallbacks; designed an **E1-anchored three-agent challenger** with direction counterfactuals and effective-sample/OOD/evidence gates for selection-adjusted E2-E5 shadow evaluation.

TECHNICAL SKILLS

Financial Engineering:
Options & Derivatives | Python & Machine Learning | C++ Programming | Advanced C++ & Modern Design